Practical – 3

AIM: Implementation of Steganography

SCOPE:

Encompasses the detection of concealed messages and data within various media files, emphasizing the application of steganographic methods for security and forensic purposes.

REQUIREMENTS:

The requirements are Hardware and Devices, Steg software, S-Tools, sample image, Documentation Templates etc.

THEORY:

Steganography:

- The root "steganos" is Greek for "hidden" or "covered," and the root "graph" is Greek for "to write." Steganography is the practice of hiding a secret message inside of (or even on top of) something that is not secret.
- Examples of steganography involve embedding a secret piece of text inside of a picture. Or hiding a secret message or script inside of a Word or Excel document. The purpose of steganography is to conceal and deceive. It is a form of covert communication and can involve the use of any medium to hide messages.
- It's not a form of cryptography, because it doesn't involve scrambling data or using a key. Instead, it is a form of data hiding and can be executed in clever ways.

Types of Steganography?

1. Text steganography

Text steganography conceals a secret message inside a piece of text. The simplest version of text steganography might use the first letter in each sentence to form the hidden message. Other text steganography techniques might include adding meaningful typos or encoding information through punctuation.

2. Image steganography

In image steganography, secret information is encoded within a digital image. This technique relies on the fact that small changes in image color or noise are very difficult to detect with the human eye. For example, one image can be concealed within another by using the least significant bits of each pixel in the image to represent the hidden image instead.

3. Video steganography

Video steganography is a more sophisticated version of image steganography that can encode entire videos. Because digital videos are represented as a sequence of consecutive images, each video frame can encode a separate image, hiding a coherent video in plain sight.

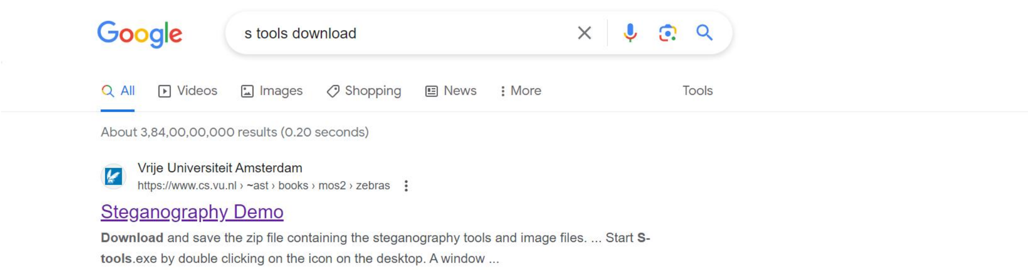
4. Audio steganography

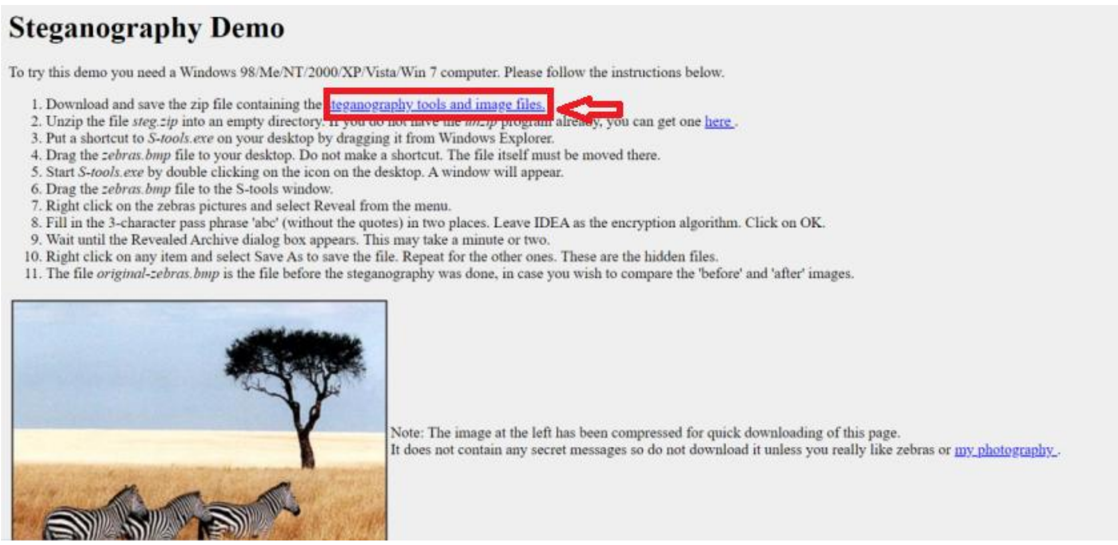
Audio files, like images and videos, can be used to conceal information. One simple form of audio steganography is "backmasking," in which secret messages are played backwards on a track (requiring the listener to play the entire track backwards). More sophisticated techniques might involve the least significant bits of each byte in the audio file, similar to image steganography.

5. Network steganography

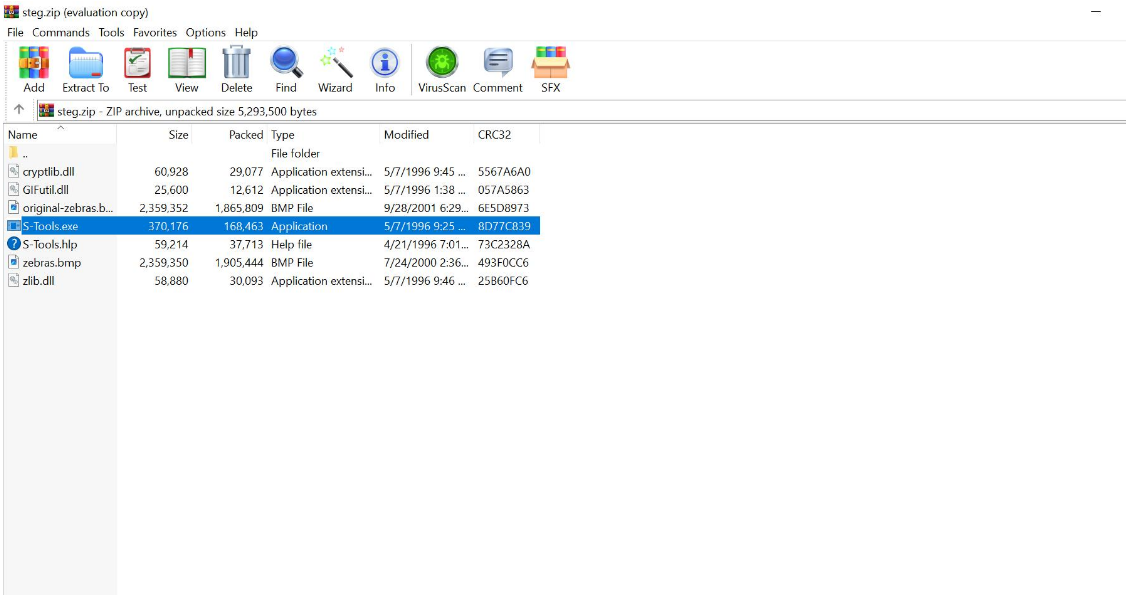
Last but not least, network steganography is a clever digital steganography technique that hides information inside network traffic. For example, data can be concealed within the TCP/IP headers or payloads of network packets. The sender can even impart information based on the time between sending different packets.

- Download Stools



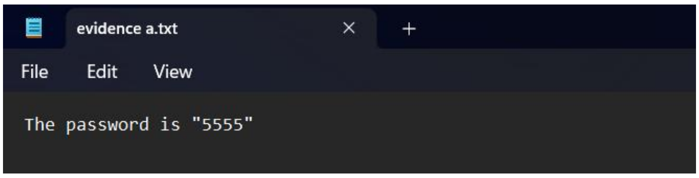


Extract the zip file in particular folder

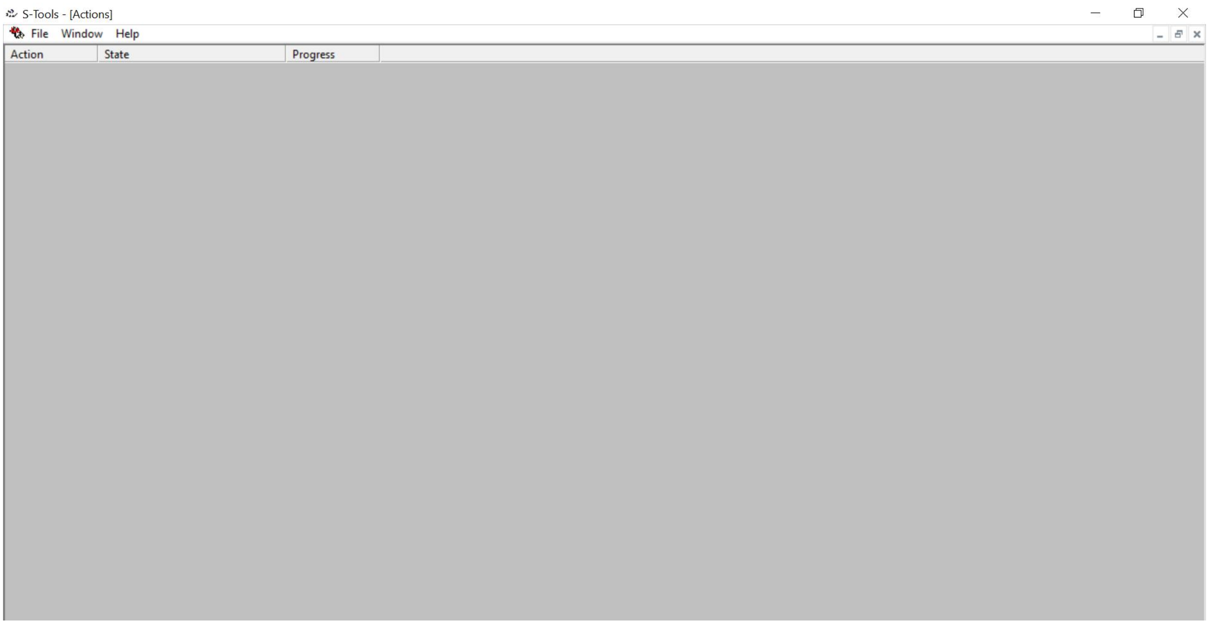


EXAMINATION:

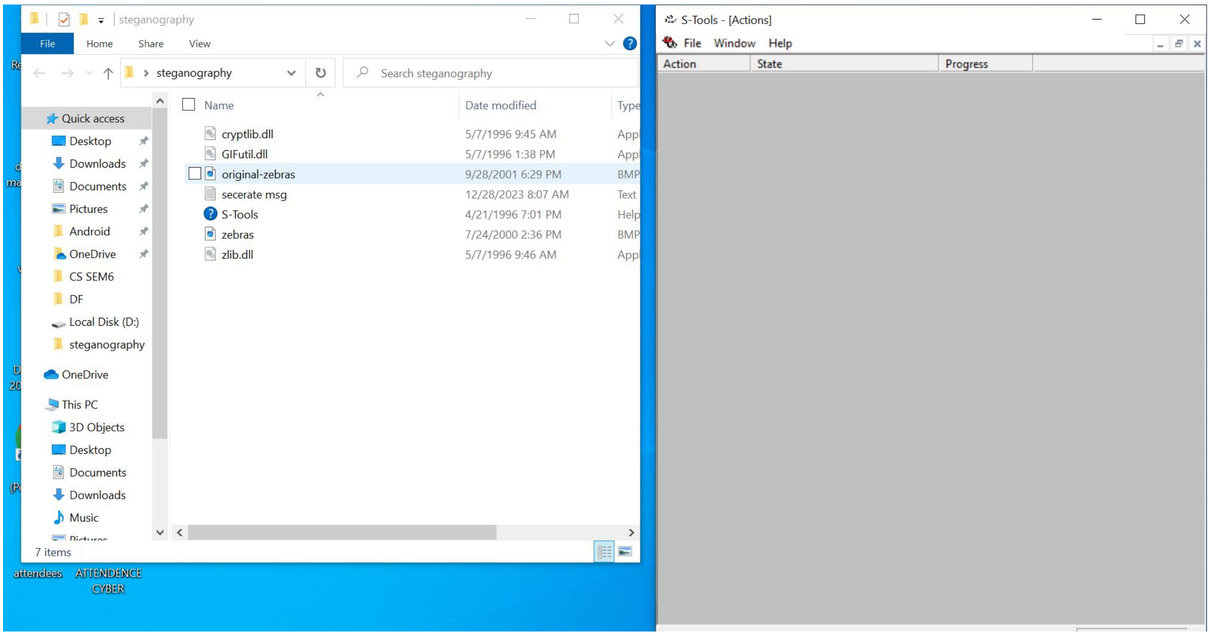
Step 1: Write a secret message in .txt format (Ex: evidence a.txt).



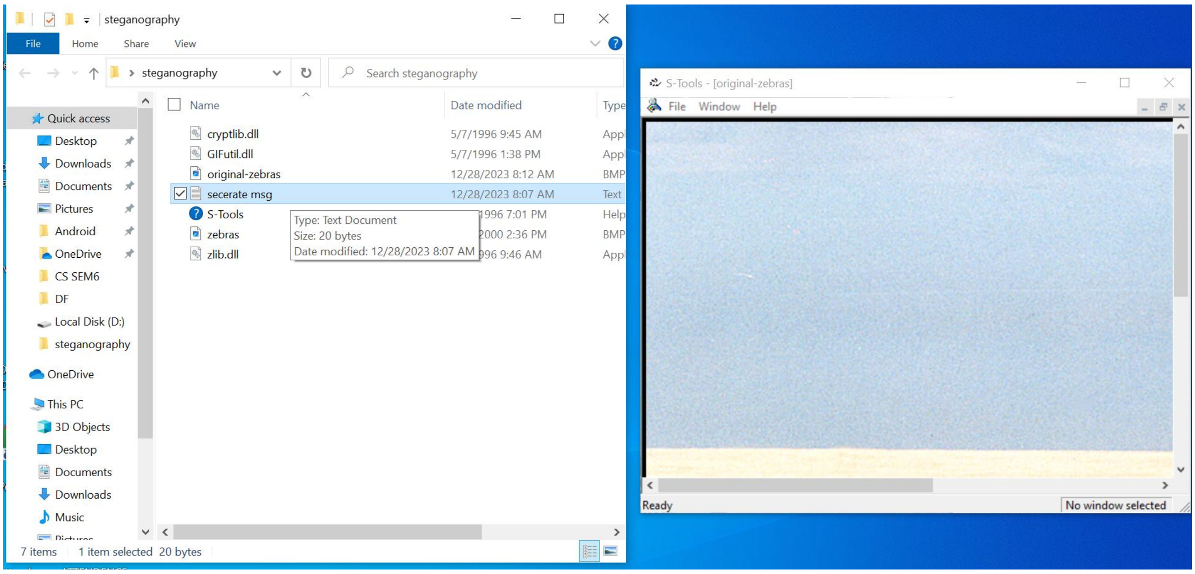
Step 2: Open the S tools



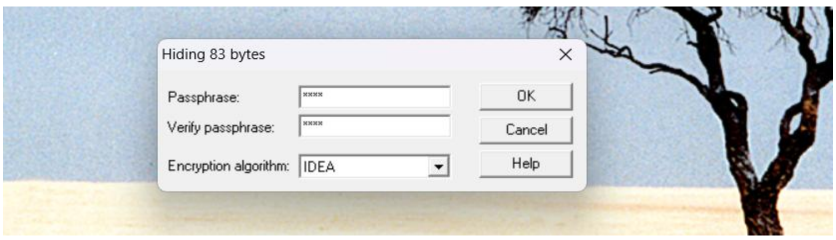
Step 3: Drag the original-zebras.bmp file and drop it into S tools



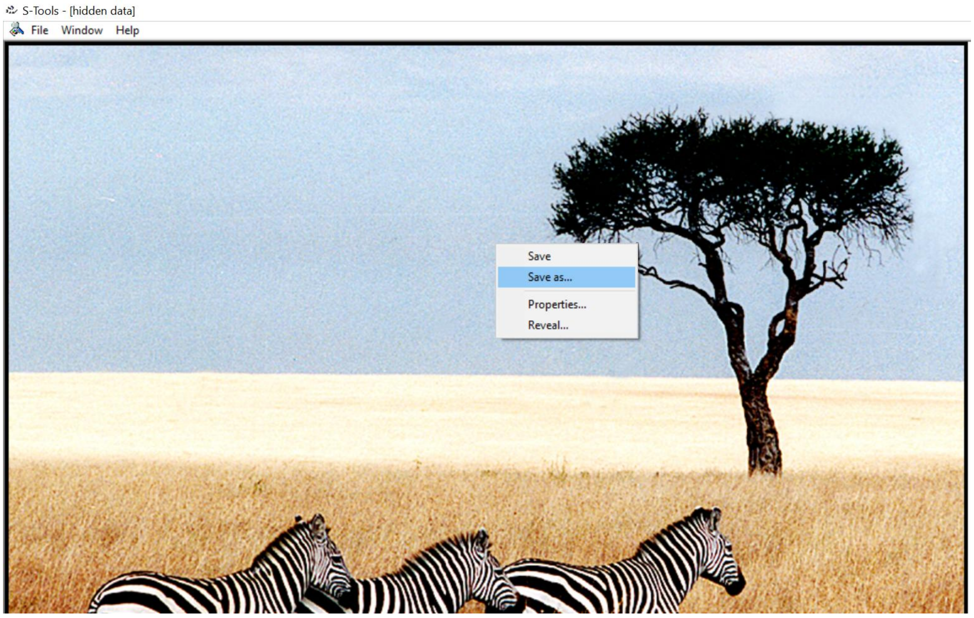
Step 4: Using S-Tools, drag and drop the secret message file on top of image file.



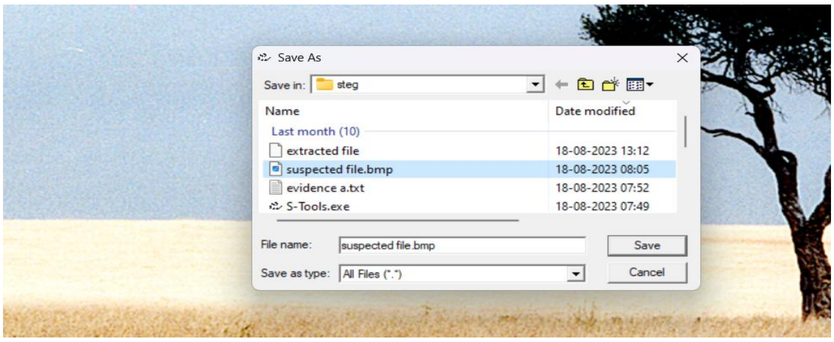
Step 5: Now to encrypt the stego file. Create a new password for the encryption and click on the “OK” button.



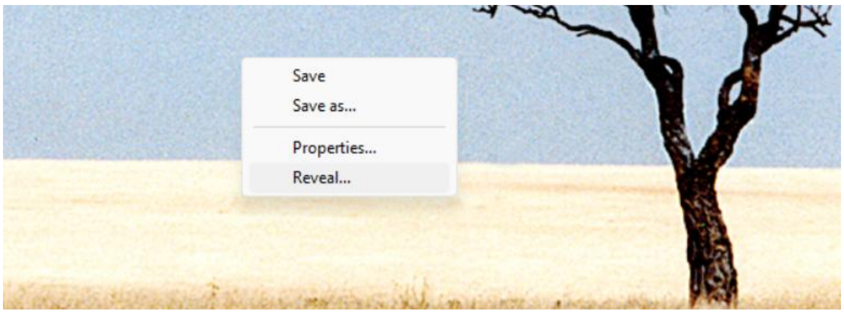
Step 6: generate a new stego image. To save the stego file, right-click on the image and select the “Save as...” option.



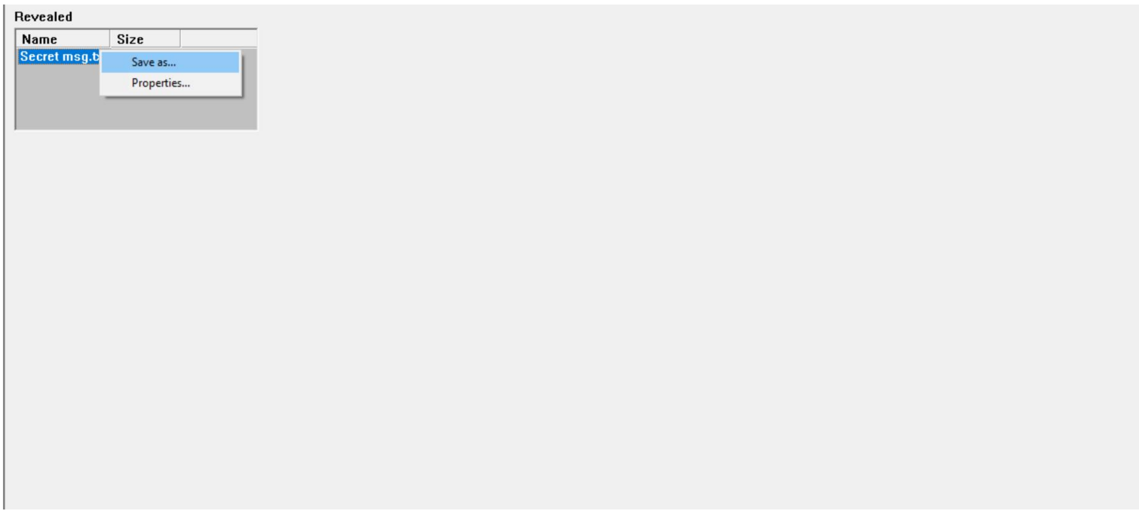
Step 7: Specify the destination to save the stego image and click on the “Save” button in order to save the file use **.BMP** extention.



Step 8: Now, to extract the concealed information from the stego image, run the S-Tools and then drag and drop suspected file into it. Right-click on the image and select the “Reveal” option from the top-down menu.



Step 9: “Revealed Archive” window displaying the secret message file name.



Step 10 : Now, Right-click on the file name and then select the "Save as…" option to save it in a location. Next check the extracted file.